3 **(a)** State the principle of superposition.

....

....

.... [2]

(b) Parallel rays of coherent light of wavelength 560 nm are incident normally in free space on two narrow slits as shown in Figure 3.1.

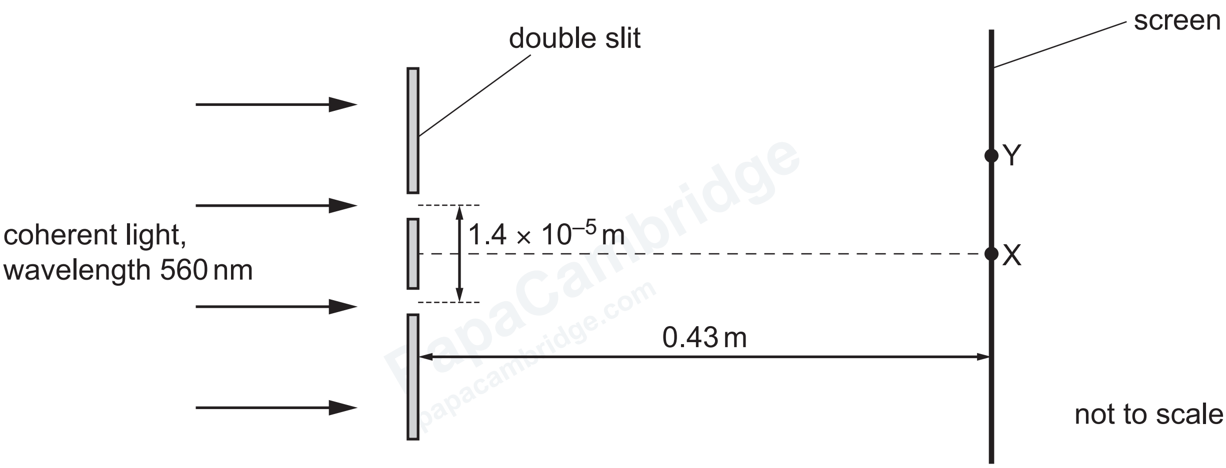

Figure 3.1

Light passing through the double slit reaches a screen that is parallel to the plane of the slits.
The distance between the centres of the slits is 1.4×10^{-5} m.
The distance between the slits and the screen is 0.43 m.

(i) State what is meant by coherent.

....

.... [1]

(ii) State what happens to the light at the point where it passes through the slits.

....

.... [1]

(iii) At point X on the screen, there is a central bright fringe.

Explain how this bright fringe is formed.

....

....

....

....

.... [3]

(iv) At point Y on the screen, the dark fringe closest to point X is formed.

Calculate the distance XY.

distance = ... m [3]

(c) On the screen in Figure 3.1:

(i) draw a cross (×) labelled with the letter B at the position of another bright fringe [1]

(ii) draw a cross (×) labelled with the letter D at the position of another dark fringe. [1]

(d) Calculate the frequency of the light in **3(b)**.

frequency = ... Hz [2]

[Total: 14]